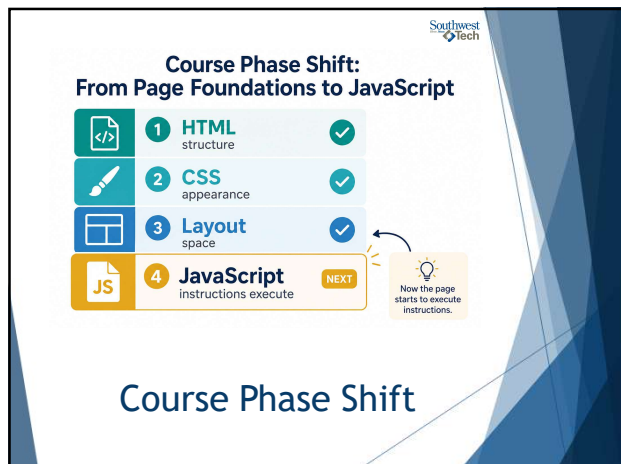




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HTML, CSS, and JavaScript: What They Do

Three parts. Different jobs. They work together to build the web.

HTML

Describes structure. Defines what content exists and how it is organized.

```
<html>
  <body>
    <h1>Welcome!</h1>
    <p>About</p>
    <p>Projects</p>
    <p>Contact</p>
  </body>
</html>
```

CSS

Describes appearance. Controls colors, fonts, spacing, and layout.

```
body {
  font-family: system-ui;
  color: #1f78b4;
}
h1 {
  background-color: #e0e0e0;
  color: white;
  padding: 10px;
}
p {
  margin-top: 10px;
  display: flex;
  gap: 10px;
}
```

JavaScript

Executes instructions. Runs step-by-step logic to perform tasks.

Example Instruction Path

- 1 Define starting values.
- 2 Check a condition.
- 3 Do something based on the result.
- 4 End when complete.

Order matters when instructions run. JavaScript follows the order you write.

Describe vs. Execute

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What Counts as Success Today

- a `.js` file runs
- output appears in the console
- variables hold values
- an `if / else` chooses a path
- you predict before you run

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What We Will Use Today

- ▶ `script.js`
- ▶ script link
- ▶ variable
- ▶ value
- ▶ Boolean
- ▶ `console.log()`
- ▶ `if / else`
- ▶ browser console

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Today's Beginner JavaScript Toolbox

Simple tools. Clear purpose. Strong foundation.

Use these tools to write, run, and see results.

- script.js**: Your JavaScript file. Where you write code.
- script link**: Connects your HTML to your JavaScript file.
- variable**: A labeled container for storing data.
- value**: The data stored in a variable.
- Boolean**: True or False. Represents yes or no.
- console.log()**: Prints information to the browser console.
- if / else**: Makes decisions and chooses a path.
- browser console**: Shows output, results, and helpful messages.

Practice: write a little, run it, and check the console. Build confidence one step at a time.

Today's Toolbox

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What We Will "Skip" For Now

- ▶ finding page elements
- ▶ button clicks
- ▶ changing visible page text
- ▶ loops
- ▶ jQuery
- ▶ full app behavior

Today: run small logic and read the result.

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Topics We'll Return To Later

Important tools. Just not today.

- DOM selection
- button clicks
- page text changes
- loops
- jQuery
- full app behavior

Not today. First, run small logic and read the result.

Parked For Later

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What A Program Does

A program is a set of instructions.
The browser runs those instructions:

from top to bottom using stored values making decisions when conditions appear producing output we can inspect

Programming begins with mental execution.

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A Program Is Step-by-Step Instructions

The computer does exactly what you tell it, in the order you tell it.

- 1 Store a value**
Save information to use later.
- 2 Print a value**
Show information as output.
- 3 Check a condition**
Ask a yes/no question.
- 4 Choose a path**
Do one thing or another based on the answer.

Programming begins with mental execution.

Program As Instructions

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Variables Store Values

A variable gives a name to a value.

Example:

```
1 const assignmentScore = 84;
2 const submittedOnTime = true;
3
```

Read it as:
"remember this value using this name."

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Console Output Makes Thinking Visible

The console is our first output space.
`console.log()` helps us see:

- ▶ what value is stored
- ▶ which line ran
- ▶ what decision happened
- ▶ whether our prediction was right

Today, console output is the visible result.

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The Browser Console: Today's Output Space

Use the console to see the results of your code.

Console Output

Console output makes thinking visible.

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Conditions Choose A Path

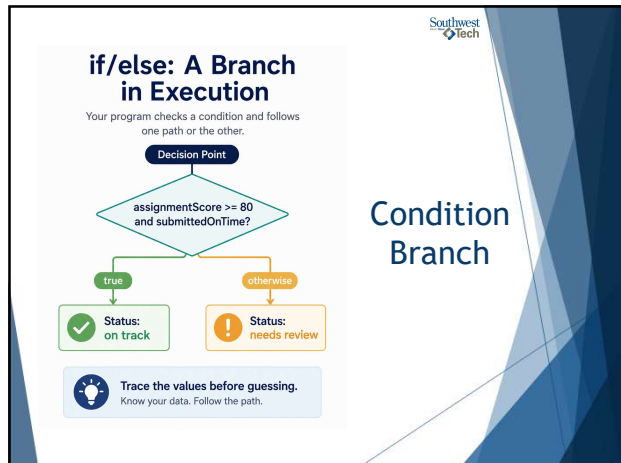
A condition creates a branch.

`if` means:
"when this is true, run this block."

`else` means:
"otherwise, run this other block."

Do not guess. -- Trace the values.

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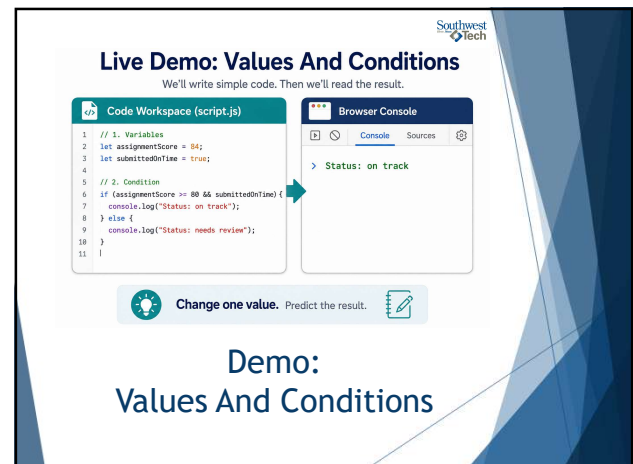
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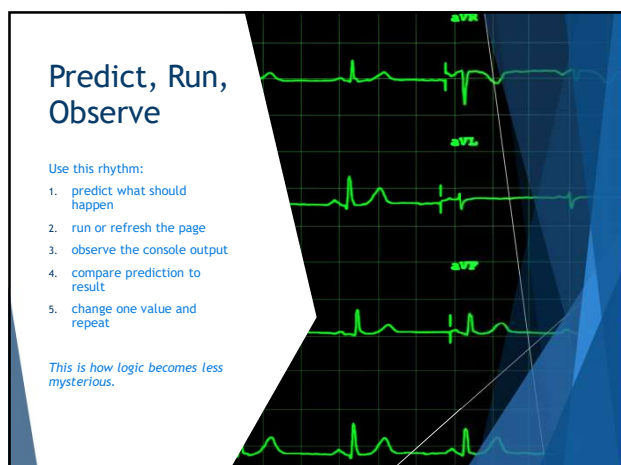
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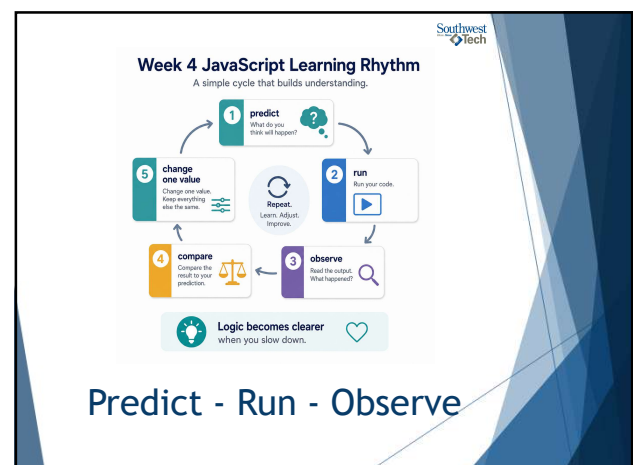
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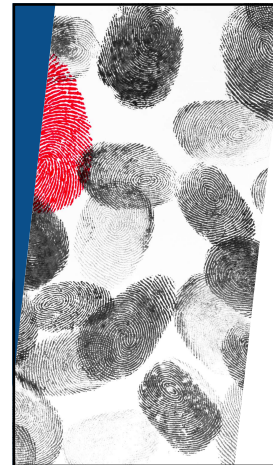
Lab

First JavaScript program
Create a `.js` file with:

- ▶ - at least 2 variables
- ▶ - at least 1 `if / else`
- ▶ - console output
- ▶ - a simple calculation or decision

*Functions are coming next;
start with values and
decisions.*

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Evidence

Preserve evidence:

- ▶ your `.js` file runs
- ▶ console output is visible
- ▶ variable names are meaningful
- ▶ the condition makes sense
- ▶ you can explain what changed
when a value changed

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Closing Success Check

Today:

- ▶ JavaScript ran as instructions
- ▶ variables stored values
- ▶ console output made logic visible
- ▶ a condition chose a path

Next:

- ▶ we organize logic into functions.

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Thank you!

Have Fun!

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